

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0695

West -Station 138 kV
February 25, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0695. The Interconnection Customer proposes a 576.9 MW natural gas combined cycle facility interconnecting at the West -Station 138 kV substation in the Maryland area. The request was received on October 28, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 301 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$78,597,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Vernon / Limon 138 KV	Easton 2 / Cady 345 KV	189.9	234	13.5
Oak Hill Substation / Lost Hills 138 KV L	Cady / Dodd Road 345 KV	188.1	215	14.5
White Bluff / Price 138 KV	Easton 2 / Cady Substation 345 KV	187.4	190	11.3
Longmire / Lost Hills 138 KV L	Cady / Dodd Road 345 KV	181.6	211	14.5
Holston Dam / Harbor 138 KV	Cady / Easton 2 345 KV line	172.7	217	9.5
Vernon / Buckskin 138 KV	Cady / Easton 2 345 KV	168.6	271	13.5
Sneca / Mckee Run 138 KV line	Cady / Dodd Road 345 KV Line	165.9	209	14.0
Buckskin / Alum Ridge 138 KV	Cady / Easton 2 345 KV	164.9	284	13.5
Cady 345/138kV Auto	Cady / Dodd Rd 345 KV L	162.7	440	22.9
Alum Ridge / Hillsboro 138 KV	Easton 2 / Cady 345 KV line	159.6	304	13.5
Longmire / Ninemile Pt 138 KV	Cady / Dodd Rd 345 KV	158.0	222	14.0
Harbor / Park Ave 138 KV L	Cady / Easton 2 345 KV Line	152.9	233	9.5
Pepper / Concord I 138 KV	N Lake / Kiwi 138 KV	139.4	180	7.8
Louisa / Cady 138 KV line	Dodd Rd / Cady 345 KV	136.1	216	9.6
Vienna Operations / Easton 2 345 KV	Linde / Ripple 138 KV Line	134.4	1,106	64.5
Louisa / Oak Hill 138 KV	Cady / Dodd Rd 345 KV	133.9	211	9.6
Southtown / Commonwealth Chesapeake 345 KV	Otay Lake Substation / Huntington Falls 138 KV	133.4	1,220	6.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Quinn / Bulverde 138 KV	Vienna Operatios / Easton 2 345 KV	129.4	273	14.1
Sooner / Concord I 138 KV L	N Lake / Kiwi 138 KV	129.0	205	7.8
Chemolite / Lammers 138 KV line	Vienna Operations / Easto 2 345 KV	127.4	217	16.7
Bozman / Spencer 138 KV	Vienna Operations / Easton 2 345 KV	126.7	279	9.7
Navy Mole / Easton Substation 138 KV L	Vienna Operations / Easton 2 345 KV	121.7	435	6.0
Warren F Sam Beasley Generation Station / Cady 138 KV	Cady / Dodd Rd 345 KV	120.4	191	7.8
Hansford / Sooner 138 KV line	Vienna Operations / Southtown 345 KV L	120.3	336	10.8
N Lake / Pepper 138 KV line	North Lake / Kiwi 138 KV	119.1	197	7.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Limon / Vernon 138 KV	Rebuild line with higher-capacity conductor	\$3,184,000	21% (\$660,000)
Oak Hill Substation / Lost Hills 138 KV L	Rebuild line with higher-capacity conductor	\$25,384,000	20% (\$5,078,000)
Price / White Bluff 138 KV L	Rebuild line with higher-capacity conductor	\$15,062,000	20% (\$3,025,000)
Longmire / Lost Hills 138 KV line	Rebuild line with higher-capacity conductor	\$14,528,000	20% (\$2,906,000)
Holston Am / Harbor 138 KV Line	Rebuild line with higher-capacity conductor	\$13,864,000	18% (\$2,464,000)
Vernon / Buckskin 138 KV	Rebuild line with higher-capacity conductor	\$6,243,000	21% (\$1,295,000)
Seneca / Mckee Run 138 KV	Rebuild line with higher-capacity conductor	\$9,545,000	19% (\$1,832,000)
Buckskin / Alum Ridge 138 KV	Rebuild line with higher-capacity conductor	\$5,400,000	21% (\$1,120,000)
Cady Substation 345/138kV TX	Replace transformer with higher-rated unit	\$16,905,000	19% (\$3,231,000)
Alum Ridge / Hllsboro 138 KV	Rebuild line with higher-capacity conductor	\$6,253,000	21% (\$1,297,000)
Ninemile Point / Longmire 138 KV line	Rebuild line with higher-capacity conductor	\$16,139,000	20% (\$3,238,000)
Harbor / Park Ave 138 KV L	Rebuild line with higher-capacity conductor	\$20,848,000	18% (\$3,705,000)
Pepper / Concord I 138 KV	Rebuild line with higher-capacity conductor	\$6,148,000	75% (\$4,613,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Louisa / Cady 138 KV L	Rebuild line with higher-capacity conductor	\$3,021,000	20% (\$591,000)
Easton 2 / Vienna Operations 345 KV Line	Rebuild line with higher-capacity conductor	\$88,064,000	19% (\$16,912,000)
Oak Hill / Louisa 138 KV	Rebuild line with higher-capacity conductor	\$2,051,000	20% (\$401,000)
Southtown / Commonwealth Chesapeake 345 KV	Rebuild line with higher-capacity conductor	\$51,674,000	2% (\$1,138,000)
Quinn / Bulverde 138 KV	Rebuild line with higher-capacity conductor	\$1,391,000	19% (\$263,000)
Concord I / Sooner 138 KV	Rebuild line with higher-capacity conductor	\$3,703,000	75% (\$2,778,000)
Lammers Substation / Chemolite 138 KV Line	Rebuild line with higher-capacity conductor	\$6,069,000	22% (\$1,344,000)
Bozman / Spencer 138 KV	Rebuild line with higher-capacity conductor	\$18,244,000	22% (\$4,069,000)
Navy Mole / Easton Substation 138 KV L	Rebuild line with higher-capacity conductor	\$8,453,000	60% (\$5,082,000)
Warren F Sam Beasley Generation Sta / Cady 138 KV	Rebuild line with higher-capacity conductor	\$25,220,000	19% (\$4,821,000)
Hansford / Sooner 138 KV line	Rebuild line with higher-capacity conductor	\$3,929,000	26% (\$1,034,000)
N Lake / Ppper 138 KV	Rebuild line with higher-capacity conductor	\$7,597,000	75% (\$5,700,000)
	Total allocated to project		\$78,597,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.